

CIVICS & COMMUNITY ENGAGEMENT

Disaster Preparedness & Community Resilience

Hazards · Risks · Vulnerability · DMC · Institutions · Citizen Action



What We Will Cover in This Module

01

Hazard vs Disaster

Defining both terms precisely and understanding the critical distinction between a natural event and a human catastrophe

02

Risks in Pakistan

The four major risk categories: floods, earthquakes, fires, and industrial/road accidents with Pakistan-specific data

03

Vulnerability & Resilience

Why some communities suffer far more than others when the same hazard strikes — and how to build resilience

04

Disaster Management Cycle

Prevention, Preparedness, Response, Recovery — what happens in each phase and who is responsible

05

Role of Institutions

Rescue 1122, hospitals, local administration, and NGOs — who does what and when in disaster response

06

Citizen Responsibilities

What every citizen **MUST** do, what they **MUST NEVER** do, and your pre-disaster home checklist

Hazard vs. Disaster: The Critical Distinction

HAZARD

A hazard is any natural or human-caused event or condition that has the potential to cause harm to people, property, or the environment. A hazard is a THREAT — it has not yet caused harm.

Hazards are neutral. They become disasters only when they interact with vulnerable human communities.

Key characteristics:

- Always present in the environment — we live with them
- Can be natural (earthquakes), technological (chemical plants), or social
- Probability and magnitude can be estimated and mapped
- Cannot always be prevented, but their effects can be reduced

Pakistan examples:

- Fault lines running through Balochistan and KPK
- Monsoon rainfall patterns over Sindh and Punjab
- Industrial facilities near residential areas in Karachi
- Steep hillsides prone to landslide in Gilgit-Baltistan

DISASTER

A disaster is a serious disruption of the functioning of a community or society that exceeds its capacity to cope using its own resources — causing widespread human, material, economic, or environmental losses.

A disaster occurs when a hazard meets a vulnerable community. The same hazard causes disaster in one place — not in another.

Key characteristics:

- Results from the interaction of a hazard with social vulnerability
- Overwhelms local capacity to respond and recover
- Requires external assistance to manage
- Scale measured by deaths, injuries, displacement, economic loss

Pakistan examples:

- The 2010 Pakistan Super Floods — 20 million displaced
- The 2005 Kashmir Earthquake — 80,000+ deaths
- The 2012 Baldia Town factory fire in Karachi — 260+ deaths
- Urban flooding in Karachi due to blocked drainage systems

The Disaster Formula and Hazard vs Disaster Comparison

The most important insight in disaster science: $\text{DISASTER} = \text{HAZARD} + \text{VULNERABILITY} - \text{CAPACITY}$. Remove vulnerability or build capacity, and the same hazard causes far less harm. This formula explains why poor communities suffer more.



	HAZARD	DISASTER
What it is	A threatening event or condition — potential harm only	Actual harm that overwhelms community capacity to cope
Requires human impact?	No — exists independently of human presence	Yes — must disrupt or overwhelm human society
Can we prevent it?	Often no — earthquakes, monsoons are unavoidable	Often yes — through preparedness and reduced vulnerability
Who determines scale?	Nature or technology determines physical magnitude	Society determines how much impact it causes by its choices
Pakistan example	A 6.5 magnitude quake beneath an uninhabited desert	A 5.0 magnitude quake beneath a dense, poorly-built city

Disaster Risks in Pakistan: Floods and Earthquakes

Pakistan is one of the world's most disaster-prone countries — ranked in the top 10 globally for climate risk. It sits on major tectonic fault lines AND faces intensifying monsoon flooding. Understanding these risks is a civic necessity for every citizen.

Floods

Key facts:

- Pakistan has experienced 24 major floods since 1950
- The 2010 super floods affected 20 million people — the single largest displacement in Pakistan's history
- Sindh, Punjab, and KPK are highest-risk provinces
- Climate change is intensifying monsoon rainfall and glacial lake outburst floods (GLOFs) in the north

Causes:

- Heavy monsoon rainfall (July–September)
- Glacial lake outburst floods (GLOFs) in Gilgit-Baltistan
- River overflow from Indus, Jhelum, Chenab, Ravi
- Poor urban drainage in Karachi, Lahore, and other cities

Impact:

- Destruction of crops and food security
- Displacement of millions into camps
- Waterborne diseases: cholera, typhoid, hepatitis
- Loss of roads, bridges, schools, and health facilities

Prevention: *Early warning (PMD) · Flood embankments · Community evacuation drills · Mangrove restoration*

Earthquakes

Key facts:

- Pakistan lies on the Eurasian-Indian plate boundary — one of the world's most seismically active regions
- The 2005 Kashmir Earthquake (7.6 magnitude) killed 80,000+ and injured 100,000+ people
- Balochistan, KPK: highest-risk zones; Quetta, Peshawar, Muzaffarabad especially vulnerable
- Small tremors occur regularly — a major earthquake is inevitable

Causes:

- Collision of the Eurasian and Indian tectonic plates
- Active fault lines: Chaman Fault, Main Karakoram, Indus-Kohistan
- Makran Coast subduction zone (tsunami risk)

Impact:

- Collapse of buildings, especially poorly-constructed ones
- Landslides blocking mountain roads and villages
- Disruption of power, water, and communications
- Long-term psychological trauma especially in children

Prevention: *Building codes (BCP 2007) · Drop-Cover-Hold drills · NDMA seismic mapping · Hospital retrofitting*

Disaster Risks in Pakistan: Fires and Accidents

While floods and earthquakes dominate disaster headlines, fires and human-caused accidents kill thousands of Pakistanis every year — and are far more preventable. These are not acts of nature. They are the result of unsafe conditions, poor regulation, and lack of enforcement.

Urban & Building Fires

Key facts:

- The 2012 Baldia Town factory fire in Karachi killed 259 workers — exits were locked from outside
- Faulty electrical wiring is the leading cause of fire in Pakistan
- Most buildings lack fire exits, alarms, or sprinkler systems
- Karachi, Lahore, and Faisalabad have among the highest fire incident rates

Main causes:

- Faulty electrical wiring — leading cause nationwide
- No fire exits, alarms, or emergency lighting
- Poor enforcement of building safety codes
- Flammable materials stored near ignition sources

Prevention: Fire inspections · Mandatory exits · Staff drills · Smoke detectors

Industrial Accidents

Key facts:

- Pakistan's chemical, textile, and construction industries have high accident rates
- Most industrial deaths occur due to absence of protective equipment (PPE)
- Child labour in hazardous industries compounds risk significantly
- Weak EOBI and Labour Department enforcement leaves workers unprotected

Main causes:

- Non-compliance with occupational safety standards
- Cost-cutting on safety equipment and training
- Inadequate emergency response procedures
- Weak third-party inspection and enforcement

Prevention: Mandatory safety training · Regular audits · PPE enforcement · Emergency drills

Road Traffic Accidents

Key facts:

- Pakistan loses 25,000–30,000 people annually to road traffic accidents
- RTAs are the leading cause of death for young people aged 15–29 in Pakistan
- Overloaded trucks, speeding buses, and motorcycle accidents are most common
- Rural roads and highway on-ramps are especially high-risk locations

Main causes:

- Driver fatigue — especially long-haul truck drivers
- Speeding, reckless overtaking, and lane violations
- Poor road conditions and absent road markings
- No helmet use on motorcycles; no seatbelt compliance

Prevention: Speed enforcement · Fatigue regulations · Road safety education · Helmet/seatbelt laws

Vulnerability and Resilience: Why Some Communities Suffer More

The same earthquake, the same flood, the same fire — and one community loses 1,000 people while another loses 10. This difference is almost never about the hazard. It is almost always about the vulnerability of the community it strikes — and how resilient that community has become.

What Makes a Community Vulnerable?

Physical Vulnerability

Poor building construction, location in flood plains or on fault lines, inadequate drainage and roads that collapse easily under disaster conditions.

Economic Vulnerability

Poverty limits the ability to prepare, evacuate, rebuild, or access healthcare. Low-income communities have no financial safety net after disaster strikes.

Social Vulnerability

Women, children, elderly, disabled persons, and minorities have less access to information, resources, and decision-making — especially during emergencies.

Institutional Vulnerability

Weak governance, poor coordination between agencies, and absent disaster plans mean communities receive inadequate protection and slow response.

Building Community Resilience

Resilience is the capacity to anticipate, withstand, respond to, and recover from disasters while maintaining core functions. Resilient communities do not just bounce back — they adapt.

Strong social networks: Communities that know and trust each other mobilise faster and share resources more effectively.

Disaster preparedness plans: Communities with plans, trained volunteers, and early warning access suffer significantly less.

Diversified livelihoods: Households with multiple income sources recover faster — not completely destroyed by single loss.

Access to healthcare & education: Healthy, educated communities make better decisions and recover more completely.

Accountable local institutions: Responsive government, accessible emergency services, and trusted leaders mean faster, fairer recovery.

The Disaster Management Cycle: Four Phases

Disaster management is a continuous cycle of four interconnected phases. Effective management is not just about responding when catastrophe strikes — the most important work happens before the disaster. Investment in prevention and preparedness saves far more lives than any response.

01

PREVENTION

Eliminate or reduce risk

Actions to eliminate or significantly reduce disaster risks before they cause harm. Targets the hazard, the vulnerability, or both.

Examples:

- Earthquake-resistant building codes
- Relocating communities from flood plains
- Fire safety regulations in factories and schools
- Deforestation control to prevent landslides

02

PREPAREDNESS

Be ready before disaster

Actions that build capacity to respond effectively when a disaster occurs — before it happens, not after.

Examples:

- Developing and practising emergency plans
- Training community first responders
- Stockpiling emergency supplies
- Establishing early warning systems

03

RESPONSE

Act when disaster strikes

Immediate actions before, during, and in the hours/days after a disaster to save lives, protect property, and meet basic human needs.

Examples:

- Search and rescue operations (Rescue 1122)
- Emergency medical treatment and triage
- Evacuation and temporary shelter
- Food, water, and sanitation distribution

04

RECOVERY

Rebuild better than before

Actions after the crisis to restore and improve community conditions — and reduce future risk through better rebuilding.

Examples:

- Psychosocial support for survivors
- Reconstruction of homes, schools, hospitals
- Restoring livelihoods: seeds, tools, livestock
- Building back better: improved construction

🔄 Continuous Cycle — Prevention saves more lives than Response

Role of Institutions in Disaster Response

No single institution can manage a major disaster alone. Effective response requires coordinated multi-agency effort. Understanding who does what prevents duplication, fills gaps, and helps every citizen know exactly who to call and what to expect.



Rescue 1122

Pakistan's primary emergency rescue service, operational in Punjab, KPK, and expanding. First responders to fires, accidents, medical emergencies, and disaster rescue operations.

What they do:

- Urban search and rescue (USAR)
- Fire fighting and emergency fire response
- Emergency medical stabilisation at the scene
- Disaster victim extraction and evacuation
- Community first aid training

Contact: Call 1122 for fires, accidents, drowning, structural collapses, or any life-threatening emergency.



Hospitals & Emergency Health

Government and private hospitals form the medical backbone. In mass casualty events, hospitals activate mass casualty plans, set up triage zones, and call in off-duty staff.

What they do:

- Triage — sorting patients by urgency of need
- Emergency surgery and intensive care
- Mass casualty incident management
- Psychosocial support for survivors
- Disease surveillance during and after floods

Contact: Go to nearest hospital or call 115 (Edhi Foundation ambulance). For life-threatening emergencies, call 1122 for transport.



Local Administration (DC/Police)

The Deputy Commissioner (DC), TMA, and Police coordinate the overall government response: evacuation orders, relief camps, law enforcement, and public communication.

What they do:

- Issuing evacuation orders and emergency declarations
- Setting up and managing relief camps
- Coordinating between agencies and preventing duplication
- Curfew and crowd control during crises
- Damage assessment reporting to NDMA/PDMA

Contact: Follow evacuation orders from the DC immediately. Do NOT return to evacuated areas without official clearance.



NGOs & Civil Society

Edhi Foundation, Al-Khidmat, Islamic Relief, and international NGOs (UN, Red Crescent) provide essential aid — often reaching communities that government cannot access quickly.

What they do:

- Emergency food, water, and non-food item distribution
- Temporary shelter and camp management
- Psychosocial support for survivors and first responders
- Community-based disaster risk reduction (CBDRR)
- Reconstruction of homes and community facilities

Contact: NGOs operate most in medium to long-term recovery and remote areas. Contact them for shelter, food, psychosocial, and livelihood support.

Citizen Responsibilities: What You MUST Do During Emergencies

In any emergency, the first responder is almost always a bystander — a neighbour, a family member, a passer-by. Professional services cannot reach everyone immediately. What you do in the first few minutes can determine who lives and who dies.

Six Things Every Citizen MUST Know and DO:

1

Stay Calm — Then Act

Panic is contagious and deadly. Take one breath, assess what you can see, and identify the most immediate threat. Calm action saves more lives than panicked movement.

When: *During: any emergency*

2

Call the Right Number Immediately

1122 (Rescue/Fire/Medical) · 15 (Police) · 115 (Edhi ambulance) · 1021 (NDMA) · 0800-26623 (PMD flood warning). Do not assume someone else has already called.

When: *During: fires, accidents, floods, medical*

3

Move to Safety First

Floods: move to high ground immediately. Earthquakes: Drop, Cover, Hold On — do not run outside during shaking. Fires: stay low, feel doors before opening, use designated exits only.

When: *During: floods, earthquakes, fires*

4

Administer Basic First Aid

Control bleeding with direct pressure. Recovery position for unconscious persons. CPR for cardiac arrest if trained. Even untrained citizens can apply pressure to wounds and keep victims calm.

When: *During: accidents, fires, collapses*

5

Follow Official Instructions

Obey evacuation orders immediately — do not delay to collect possessions. Follow Rescue 1122 and police instructions at the scene. Tune to Radio Pakistan, ARY, or Geo for official updates.

When: *During: floods, major disasters*

6

Help Vulnerable People First

Assist elderly, disabled, and injured persons in your immediate area if you are safe to do so. Knock on neighbours' doors during evacuations. Never leave vulnerable persons without notifying emergency services.

When: *During: all emergencies*

What You Must NEVER Do — and Your Pre-Disaster Home Checklist

Six Actions That Cost Lives — Never Do These:

✗ Never re-enter a flooded or fire-damaged building

Structural damage may not be visible. Floods can trap you; fire can reignite. Wait for official clearance.

✗ Never drive through floodwater

60cm of water can sweep away a car. More people drown in vehicles than anywhere else during floods. Turn around.

✗ Never spread unverified information on social media

False rumours cause mass panic, misdirect resources, and delay rescue. Verify before you share.

✗ Never ignore official evacuation orders

'I know my area' is the most common last thought of flood victims. Officials have data you don't. Evacuate.

✗ Never interfere with Rescue 1122 operations

Crowds slow rescue, obstruct equipment, cause secondary accidents. Keep clear and direct others away.

✗ Never use elevators during fires or earthquakes

Elevators may lose power or fail structurally. Always use stairs for fire evacuation or after a quake.

Pre-Disaster Home Checklist

✓ Know your risks

- What hazards are most likely in your area?
- Is your building structurally sound?
- Where is your nearest relief camp?

✓ Emergency contacts

- 1122 Rescue · 15 Police · 115 Edhi
- Save local hospital emergency number
- Know your immediate neighbours' numbers

✓ Emergency kit

- 3-day water supply (3L per person per day)
- First aid kit and essential medications
- Torch, batteries, whistle, and cash

✓ Family plan

- Agree on a meeting point outside your home
- Identify a contact person outside your city
- Practice your fire evacuation route now

KEY TAKEAWAYS

Disaster Preparedness & Community Resilience

Hazard vs Disaster

A hazard is a potential threat. A disaster = Hazard + Vulnerability – Capacity. The same earthquake kills 80,000 in one city and 80 in another — the difference is community vulnerability and preparedness.

Pakistan's Risks

Floods (worst in Sindh/Punjab), earthquakes (KPK/Balochistan), fires (urban/factory), and road accidents (25,000+ deaths/year). These risks are intensifying with climate change.

Vulnerability

Poor, marginalised, and physically exposed communities suffer most. Resilience is built through preparedness, strong social networks, diverse livelihoods, and accountable institutions.

DMC

Prevention → Preparedness → Response → Recovery — a continuous cycle. Prevention saves more lives than any response can. Investment before a disaster always costs less than recovery after.

Institutions

1122 for rescue/fire/medical emergencies. Hospitals for treatment. DC/Police for coordination. NGOs for aid and recovery. Know who to call — do not assume someone else already has.

Citizen Action

Stay calm. Move to safety. Call 1122. Help vulnerable people first. Follow official instructions. NEVER re-enter flooded buildings, drive through floodwater, or spread unverified rumours.